

Inventory & Supply Chain Management Analysis

1-Project Overview:

The Inventory & Supply Chain Management Analysis project is an interactive Power BI dashboard designed to analyze inventory performance, warehouse utilization, sales activity, transportation costs, lead times, and backorders.

The dashboard transforms inventory and supply chain data into meaningful business insights, helping identify operational inefficiencies, monitor inventory levels, evaluate warehouse capacity, and understand regional and category-level performance.

The analysis focuses on daily inventory and sales activity while providing interactive filters for Region and Category, allowing users to explore supply chain performance from different business perspectives.

2-Dataset Summary:

Rows: 1200

Columns: 15

Key Features:

Columns name (Date, Region, Category, Supplier, Warehouse, Order Status, Units Sold, Inventory Level, Transportation Cost, Order Accuracy, Lead Time, Backorder, Cost of Goods Sold, Average Inventory, Warehouse Capacity)

3. Business Problem

Inventory and supply chain operations involve several interconnected areas, including sales, inventory levels, warehouse capacity, transportation, suppliers, order fulfillment, and backorders.

Without a centralized analytical dashboard, it can be difficult for management to understand:

- Whether inventory is being efficiently managed
- How much warehouse capacity is being utilized
- Which regions have higher transportation costs
- Which categories have longer lead times
- How sales are changing over time
- Where backorders are occurring

- How inventory is distributed across regions and categories

4. Project Objectives

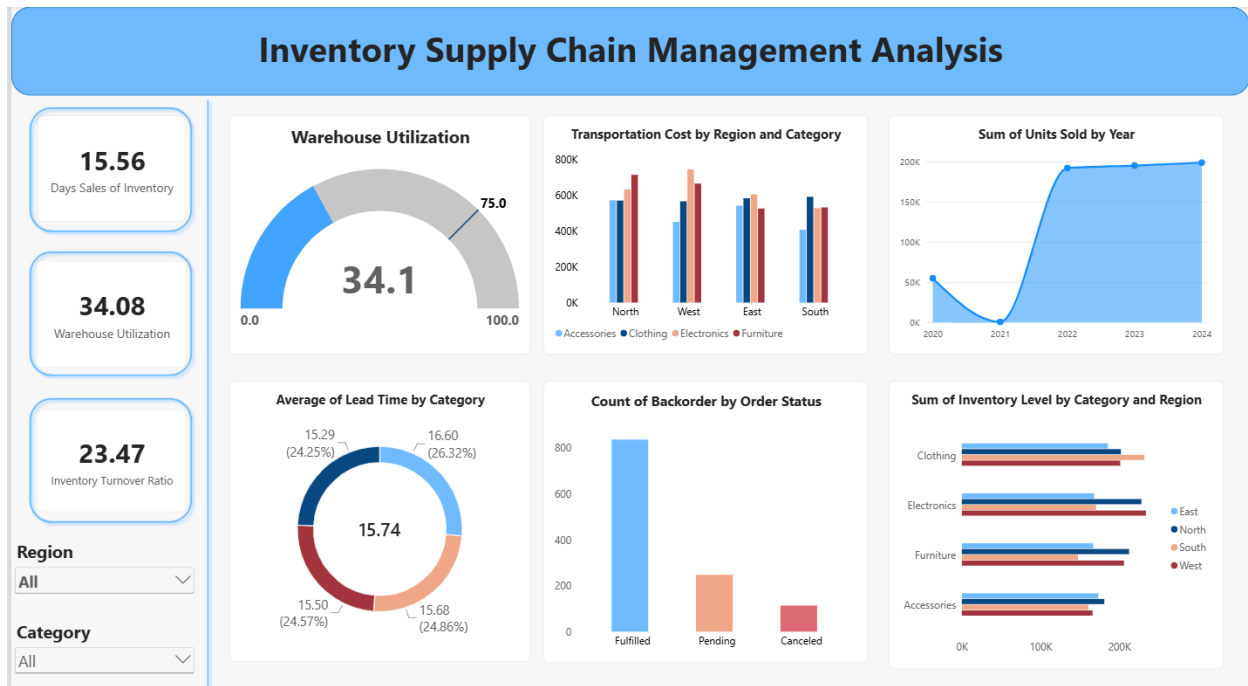
- Analyze inventory and sales performance.
- Monitor warehouse utilization.
- Measure inventory turnover.
- Analyze transportation costs.
- Identify yearly sales trends.
- Compare average lead time by category.
- Analyze backorders by order status.
- Analyze inventory distribution by category and region.
- Provide interactive Region and Category filtering.
- Create a centralized dashboard for supply chain analysis.

5. Tools Used

Microsoft Power BI

- Power Query
- Data Modeling
- Data Visualization
- Interactive Dashboard

6. Dashboard Using Power BI:



7. Final Dashboard Insights

You can use this section almost directly:

Key Findings

1. Warehouse utilization is approximately 34%, which is below the dashboard's displayed 75% reference level.
2. Inventory turnover is approximately 23.47, indicating substantial inventory movement relative to average inventory.
3. Days Sales of Inventory is approximately 15.56 days.
4. Unit sales show a major increase after 2021 and relatively stable growth from 2022 onward.
5. Transportation costs vary across both regions and product categories.
6. Average category lead times are generally around 15–17 days.
7. Backorder-related records are highest under the Fulfilled status in the current visualization.

8. Inventory levels differ across category-region combinations, providing opportunities for better inventory allocation.

8. Recommendations

Recommendation 1 — Warehouse Optimization

Since the dashboard shows utilization below the displayed reference level, management could evaluate whether warehouse space is being efficiently allocated.

Recommendation 2 — Transportation Optimization

Investigate regions/categories with higher transportation costs and evaluate:

- Delivery routes
- Supplier locations
- Shipment consolidation
- Transportation providers

Recommendation 3 — Inventory Optimization

Use category and region-level inventory information to avoid both:

- Overstocking
- Stockouts

Recommendation 4 — Lead Time Reduction

Investigate categories with higher average lead times and identify supplier or logistics bottlenecks.

Recommendation 5 — Backorder Monitoring

Track actual backordered quantities rather than simply counting backorder records if the Backorder column contains numeric quantities.

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